

```

8   userId: string;
9   title: string;
10  message: string;
11  type: 'info' | 'success' | 'warning' | 'error';
12  read: boolean;
13  createdAt: any;
14  link?: string;
15 }
16
17 export async function createNotification(
18   data: Omit<AppNotification, 'id' | 'read' | 'createdAt'>
19 ): Promise<string> {
20   const ref = await addDoc(collection(db, 'notifications'), {
21     ...data,
22     read: false,
23     createdAt: serverTimestamp(),
24   });
25   return ref.id;
26 }
27
28 export async function getUnreadCount(
29   userId: string
30 ): Promise<number> {
31   const q = query(
32     collection(db, 'notifications'),
33     where('userId', '==', userId),
34     where('read', '==', false)
35   );
36   const snapshot = await getCountFromServer(q);
37   return snapshot.data().count;
38 }
39
40 export async function markAsRead(
41   notificationId: string
42 ): Promise<void> {
43   await updateDoc(doc(db, 'notifications', notificationId), {
44     read: true,
45   });
46 }

```

Listing 8.8: Notification Service (notification-service.ts)

Notifications are stored as Firestore documents, enabling real-time synchronization. The `getUnreadCount`